Frequent Itemset Generation (Min Support = 40%)

Total Transactions = 7
Minimum Support = 40% = 3 transactions

Transactions:
T1: Bread, Milk, Chips, Mustard
T2: Beer, Diaper, Bread, Eggs
T3: Beer, Coke, Diaper, Milk
T4: Beer, Bread, Diaper, Milk, Chips
T5: Coke, Bread, Diaper, Milk
T6: Beer, Bread, Diaper, Milk, Mustard
T7: Coke, Bread, Diaper, Milk

Step 1: Frequent 1-Itemsets

Item	Count	Support
Bread	6	85.7%
Milk	6	85.7%
Diaper	6	85.7%
Beer	4	57.1%
Coke	3	42.9%

Items appearing less than 3 times were removed (Chips, Mustard, Eggs).

Step 2: Frequent 2-Itemsets

Itemset	Count	Support
{Bread, Milk}	5	71.4%
{Bread, Diaper}	5	71.4%
{Milk, Diaper}	5	71.4%
{Beer, Diaper}	4	57.1%
{Bread, Beer}	3	42.9%
{Milk, Beer}	3	42.9%
{Milk, Coke}	3	42.9%

{Diaper, Coke}	3	42.9%

Step 3: Frequent 3-Itemsets

Itemset	Count	Support
{Bread, Milk, Diaper}	4	57.1%
{Bread, Beer, Diaper}	3	42.9%
{Milk, Beer, Diaper}	3	42.9%
{Milk, Diaper, Coke}	3	42.9%

Step 4: 4-Itemsets

Possible combinations from 5 items = C(5,4) = 5, but none appeared at least 3 times. Therefore no frequent 4-itemsets.

however

Frequent 1-itemsets: Bread, Milk, Diaper, Beer, Coke
Frequent 2-itemsets: 8 sets
Frequent 3-itemsets: 4 sets
Frequent 4-itemsets: None

Apriori principle was used: any set containing an infrequent subset is removed.